

Three Methods to Compare e^π and π^e

Sun Shengran

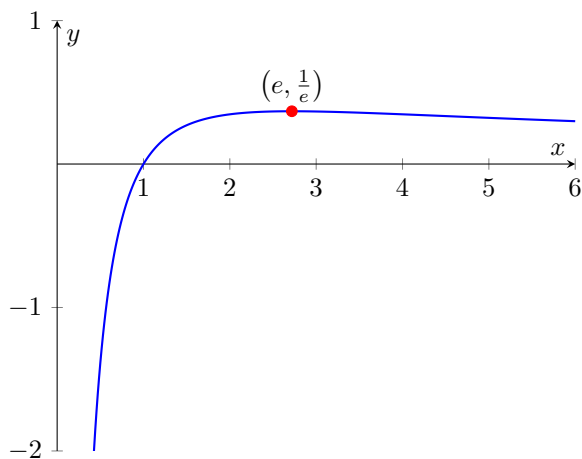
November 2025

1 Introduction

It is interesting to combine two essential constants into two numbers and work out some relationships between them. Thus, in this article, I will discuss three methods to compare them. The first two methods both relate to logarithm and the last one is more elegant.

2 First Method: Take Logarithm Both Sides

2.1 Lemma: Graph of $y = \frac{\ln(x)}{x}$, $x \in \mathbb{R}^+$



2.2 Main Process of Comparison

$$e^\pi \text{ v.s. } \pi^e$$

$$\Leftrightarrow \pi \ln(e) \text{ v.s. } e \ln(\pi)$$

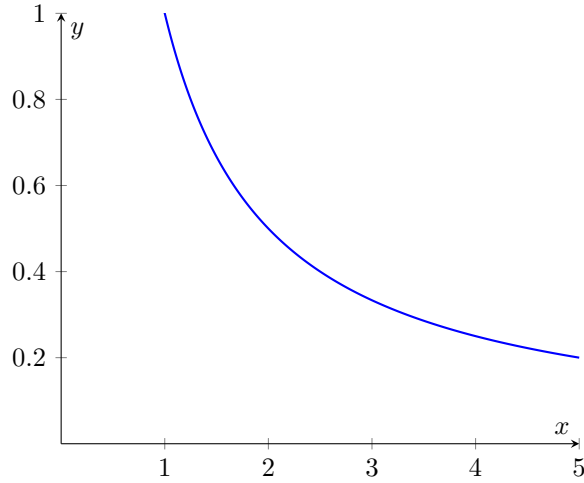
(take natural logarithm as logarithm is a strictly increasing function)

$$\Leftrightarrow \frac{1}{e} \text{ v.s. } \frac{\ln(\pi)}{\pi}$$

By lemma 2.1, $\frac{1}{e}$ is the maximum of the function $\frac{\ln(x)}{x}$. Hence, $e^\pi > \pi^e$.

3 Second Method: A Classic Area Comparison

Considering the graph of $y = \frac{1}{x}$, $x > 0$:



The area of the rectangle with width $(\pi - e)$ and height $\frac{1}{e}$ is larger than the area enclosed by the curve $y = \frac{1}{x}$ and x-axis between $x = e$ and $x = \pi$.

Hence,

$$\begin{aligned} \frac{\pi-e}{e} &> \int_e^\pi \frac{1}{x} dx \\ \Rightarrow \frac{\pi}{e} - 1 &> \ln(\pi) - 1 \\ \Rightarrow \pi &> \ln(\pi^e) \\ \Rightarrow e^\pi &> \pi^e. \end{aligned}$$

4 Third Method: Applying $e^x \geq x + 1$

4.1 Lemma: $e^x \geq x + 1$, attaining equal sign iff $x = 0$

This can be simply inspected from the graphs of $y = e^x$ and $y = x + 1$ on the same axes.

4.2 Main Process of Comparison

$$e^\pi = e^{e(\frac{\pi}{e})} = (e^{(\frac{\pi}{e}-1+1)})^e = (e^{(\frac{\pi}{e}-1)})^e e^e \geq (\frac{\pi}{e})^e e^e = \pi^e$$

as $\frac{\pi}{e} - 1 \neq 0$, $e^\pi > \pi^e$.

5 Summary

The first method involves taking logarithm which is the most common approach to tackle exponents. Using the graph of $y = \frac{\ln(x)}{x}$ which we are familiar with is actually a kind of symmetry exploitation.

In comparison, the second method seems to be not that natural and not easy notice. Although this approach, relying on the area under the curve $y = \frac{1}{x}$ is quite common, it is not simple to come up with it immediately.

The third method is more competition-styled. The application of the inequality $e^x \geq x + 1$ is not uncommon in mathematics competition. Even so, the key point of this approach is the algebraic manipulation that is incredibly grateful and skillful.

6 Reference

1. Second Method Source: A level Mathematics B (MEI), October 2021, OCR